CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

REVISED TOPIC

AI-powered Quality Control and Defect Detection in Electronics Assembly

Reducing false calls of automated optical inspection (AOI) with Siemens production data

Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

Team 1 · Guided by Dr. Hamidreza Heidari · 23 October 2026 · github.com/bodhi584/aoi-decision-gate

INTRODUCTION

Too many false alarms after automated inspection

98 %

of the boards flagged by the AOI machine are fine (false calls) in real Siemens production data.

Every flagged board goes to a manual inspection station.

Based on the lecture example (p. 18)

“Siemens – Quality Control and Defect Detection”. Key motivations: reduce dependence on manual quality checks, keep product quality, increase efficiency.

A current problem: Siemens and NVIDIA, Erlangen, 21 Oct 2026

AI second check of AOI results: real defect or false call? Target: about 40 % fewer false calls.

Our goal

Classify each AOI alarm as true defect or false call: fewer false calls to manual inspection, no extra defects missed.

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ONE GOAL: FEWER FALSE CALLS, WITHOUT MISSING DEFECTS

Input → System → Output

INPUT

AOI alarm

flagged board: about 70 measurements per solder joint

component type, AOI error code

SYSTEM (OURS)

Binary classification

supervised machine learning, trained on past manual-inspection results

threshold: ≤ 1 % of defects missed

OUTPUT: IS THE AOI ALARM REAL?

False call

→ skip manual inspection

True defect

→ manual inspection, repair

Key factors

False calls

each one costs operator time at the manual station

Missed defects

a missed defective board reaches the customer

Changes over time

the line changes; a model must keep up

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PROJECT SCOPE

What we will and will not do

Aim

This project aims to reduce false calls after automated optical inspection: a binary classifier decides for each AOI alarm whether it is a true defect or a false call, while at most 1 % of defects are missed.

Includes

- Data preprocessing: Siemens AOI data
- Training three models
- Day-by-day simulation of the line
- Evaluation using standard metrics

Out of scope (current phase)

- Real hardware deployment
- Integration with physical AOI machines
- Changing the AOI camera images

Conditions and limitations

- One Siemens line, 132 days
- Measurements anonymised (no physical names)
- Labels from operators may contain errors
- Licence: non-commercial (course use)

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What the project will achieve, and how we measure it

1. Reduce false calls: train three classifiers to separate false calls from true defects.
2. Keep quality: set the classification threshold so that at most 1 % of defective boards are missed.
3. Test on the line: integrate the models into a day-by-day simulation of the Siemens line.
4. Compare the three models and identify the most suitable one.

Requirements (two KPIs)

≥ 20 % false calls correctly identified, i.e. manual inspections saved (baseline 6.4 %)

≤ 1 % defective boards missed (classified as false call)

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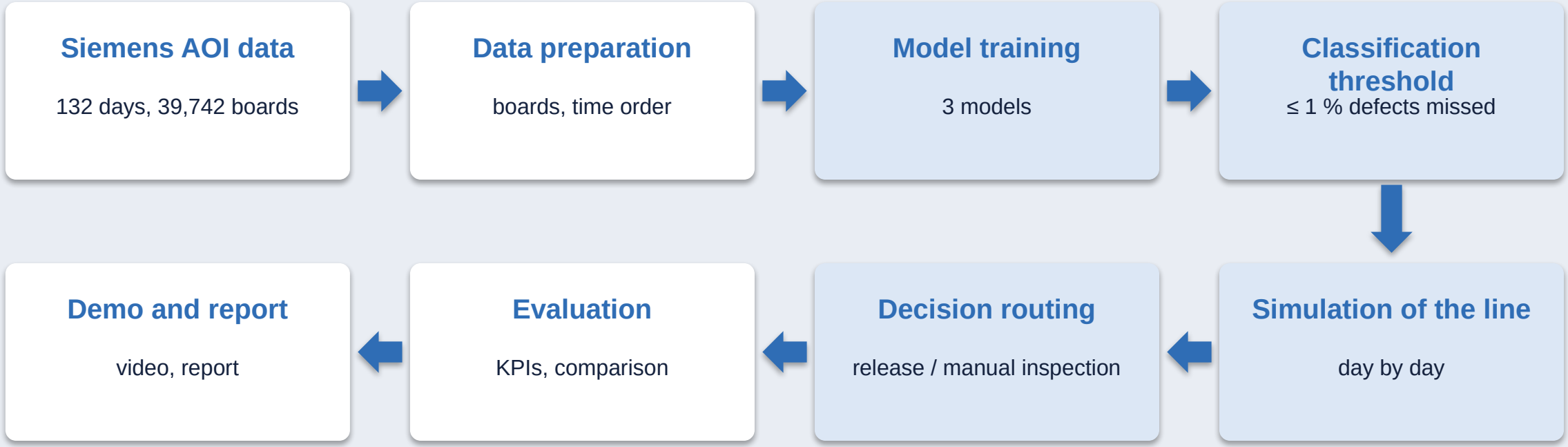
TECHNOLOGY STACK

Data science tools from the lecture (p. 22)

Process	Tool / Framework	Purpose
Data preparation	Python, pandas	Load the Siemens AOI data, group joints into boards, keep time order
Model training	scikit-learn	Logistic regression, random forest, neural network (one hidden layer)
Simulation	Python	Day-by-day simulation of the line: release or manual inspection
Evaluation	scikit-learn, pandas	Precision, recall, confusion matrix, missed defects
Visualization	matplotlib, Jupyter	Graphical result representation, step-by-step notebook
Code sharing and tests	GitHub, pytest	Repository, tests on every change

PROJECT FLOW CHART

From the Siemens data to the KPIs



Blue boxes: the system we build. White boxes: data and project steps.

WORKING METHODOLOGY

CRISP-DM in four steps

Step 1 · Data preparation

Collected Siemens AOI data: flagged boards, defective & false call

Grouped solder joints into boards; encoded categories

Split data in time order into train, validate, test sets

Step 2 · Model development

Trained logistic regression as baseline

Trained random forest and a neural network (one hidden layer)

Set the classification threshold: ≤ 1 % of defects missed

Step 3 · Simulation integration

Simulated the line day by day

Each day decided by a model trained on earlier days only

Boards are virtually routed to release or manual inspection

Step 4 · Evaluation and comparison

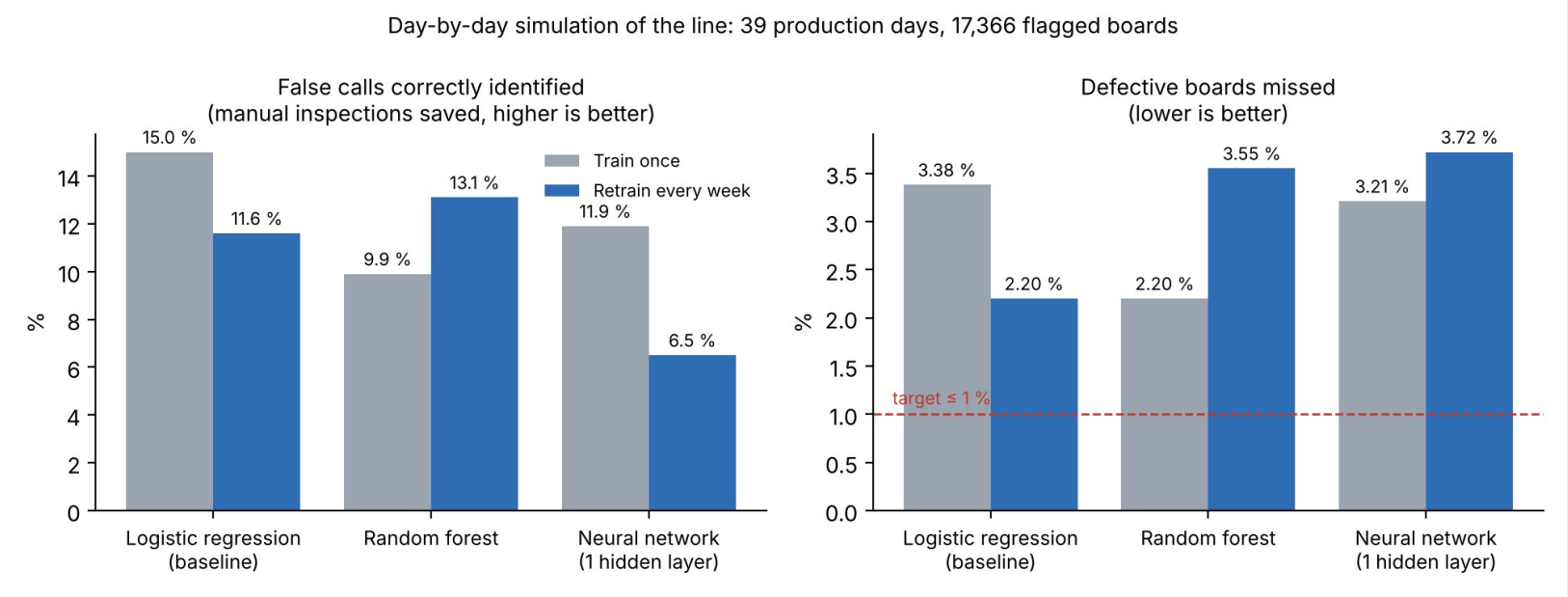
Analyzed precision, recall, missed defects, inspections saved

Compared the three models to identify the best-performing one

Error analysis; demo video and report

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Day-by-day simulation of the line: three models running



What it shows

Today: 6–15 % of false calls correctly identified (manual inspections saved).

But 2.2–3.7 % of defective boards are missed: above the 1 % limit.

Next: ≥ 20 % identified at ≤ 1 % missed.

BILL OF MATERIALS (BOM)

What the project needs

Component	Specification	Quantity	Cost
Data set	Siemens AOI data, Mendeley (CC BY-NC 3.0), 333 MB	1	Free
Computer	Normal laptop or Google Colab (no graphics card needed)	4	Free
Software	Python and open-source libraries	–	Free
Code hosting	GitHub	1 repository	Free

Total cost: 0 €

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EXPECTED OUTCOMES

What we hand in

- ✓ Successfully trained binary classifier for AOI alarms (Python software prototype)
- ✓ Functional day-by-day simulation of the line with decision routing
- ✓ Comparative performance metrics for the three models
- ✓ Identification of the most suitable model, with limits and error analysis
- ✓ Demo video, run instructions, report, public GitHub repository

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SAFETY & ETHICAL CHECKLIST

Keeping it safe and realistic

- ✔ Simulation-only environment; no physical machinery risk
- ✔ Dataset is anonymized by Siemens (no personal data); used under its licence
- ✔ Ethical AI principles followed: transparency, fairness, accountability
- ✔ Avoid dataset bias (time-ordered split) and ensure reproducibility of results
- ✔ Safety first in the design: limit on missed defects (≤ 1 %)

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FUTURE WORK

From the public data set to a real line

1 · Industry partner

An electronics plant shares AOI and manual-inspection records.

2 · AOI interface

Read each board’s AOI result from the machine; send the decision to the repair station.

3 · Shadow mode

The classifier runs next to the operators and only logs its results.

4 · Approval

Quality engineers compare the logs before any board skips manual inspection.

BIBLIOGRAPHY

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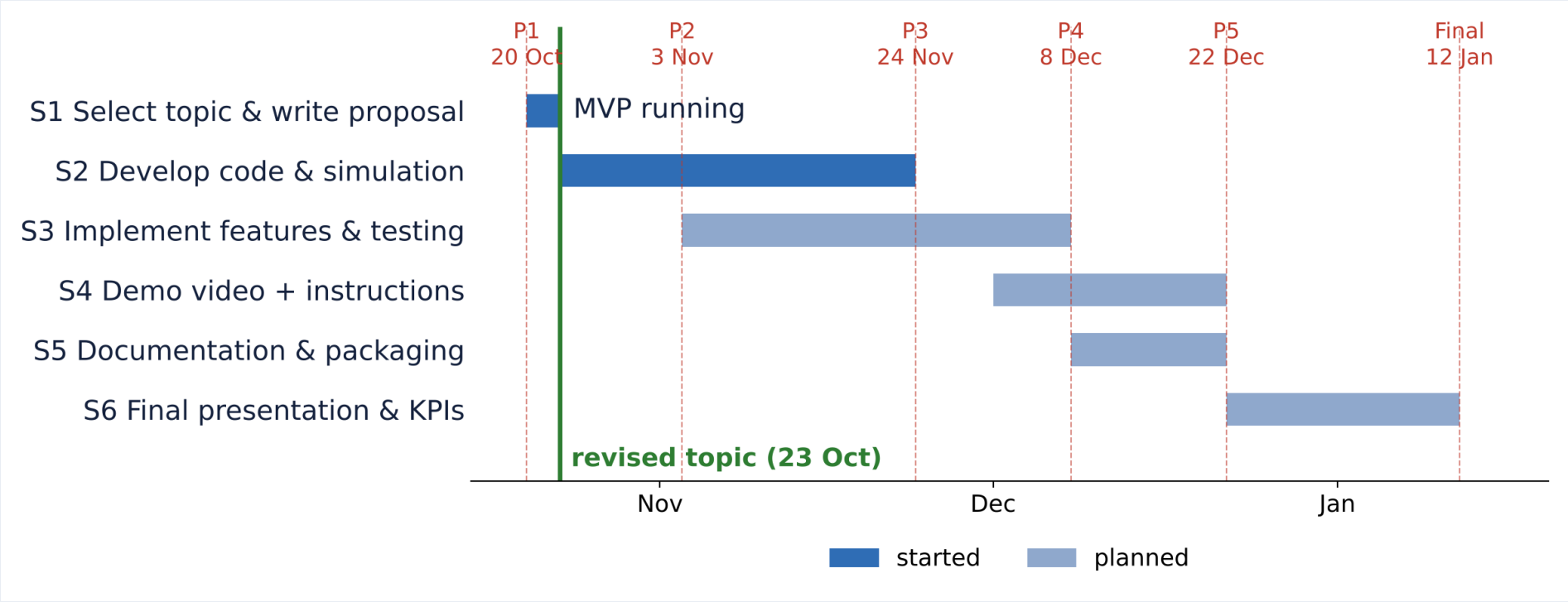
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Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

Chapman, P. et al. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS.

Lecture slides, Case Study Intelligent Systems in Production, p. 8, 12, 18, 22, 33.

Software & Simulation project, S1–S6 (lecture p. 8, 33)



Repository link

github.com/bodhi584/aoi-decision-gate

Thank you

Team 1 · Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas